

TEST CASE DOCUMENT

Smart Student Productivity Platform

1. Document Control

Field	Details
Document Title	Test Case Document
Project Name	Smart Student Productivity Platform
Organization	SR Pvt Ltd
Document Version	1.0
Document Status	Draft
Prepared By	Tech Lead
Testing Type	Functional, Integration, Security, UAT and Non-Functional Testing

2. Revision History

Version	Date	Author	Description	Status
1.0	24/09/2026	Tech lead	Initial Test Case Document	Draft

3. Approval

Role	Name	Signature	Date
Project Manager / BA	Sakshi Shingole		
Scrum Master	Sweta Kadam		
Tech Lead	Riddhi Zunjarrao		
Solution Architect	Ragini Singh		

4. Introduction

The Smart Student Productivity Platform is a centralized academic web application designed to manage academic activities and productivity for students, faculty members, HODs/domain administrators, and super administrators.

The platform provides academic management, assignments, examinations, attendance, learning resources, communication, online classes, AI-based doubt assistance, study groups, notifications, reporting, and administrative management.

The purpose of this Test Case Document is to define the test scenarios and test cases required to verify that the implemented system meets the approved functional and non-functional requirements.

5. Testing Objectives

The objectives of testing are:

1. Verify that all major system functionalities work as expected.
 2. Verify authentication and authorization.
 3. Verify role-based access control.
 4. Verify Student Portal functionality.
 5. Verify Faculty Portal functionality.
 6. Verify HOD/Domain Admin functionality.
 7. Verify Super Admin functionality.
 8. Verify academic data relationships.
 9. Verify assignment and assessment workflows.
 10. Verify attendance and result management.
 11. Verify AI Doubt Solver functionality.
 12. Verify study group functionality.
 13. Verify notifications and communication.
 14. Verify file upload and storage functionality.
 15. Verify security controls.
 16. Verify integration between frontend, backend, Firebase and external services.
 17. Verify error handling and validation.
 18. Verify responsive and usable interfaces.
 19. Verify critical business rules.
 20. Provide evidence for UAT and final acceptance.
-

6. Scope of Testing

6.1 In Scope

The following modules are included:

- Login and authentication
- Role identification
- Authorization
- Student Portal
- Faculty Portal
- HOD Portal
- Super Admin Portal
- Academic structure
- Courses and subjects
- Assignments
- Assessments
- Projects
- Examinations
- Attendance
- Results
- Learning materials
- Tasks
- Calendar
- Communication
- Online classes
- Notifications
- AI Doubt Solver
- Faculty doubt escalation
- Study groups
- Exam-focused study
- User management
- Role and permission management
- Reports
- Audit logs
- File management
- Firebase integration
- Cloudinary integration
- Cloud Functions
- Security Rules
- FCM notifications
- Google Meet integration where implemented

6.2 Out of Scope

Any functionality not defined in the approved BRD/SRS is outside the scope of this test document.

8. User Roles

Role	Main Responsibility
Student	Access academic information and productivity features
Faculty	Manage assigned academic activities
HOD / Domain Admin	Manage department/domain academic activities
Super Admin	Manage platform-wide users, structure and configuration

11. Authentication Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-AUTH-001	FR-AUTH-001	Login with valid student credentials	Valid student email/password	1. Open login page. 2. Enter valid email. 3. Enter valid password. 4. Click Login.	User is authenticated successfully and redirected to the Student Portal/Dashboard.	Critical
TC-AUTH-002	FR-AUTH-001	Login with invalid password	Valid email + incorrect password	1. Enter valid email. 2. Enter incorrect password. 3. Click Login.	Login is rejected and an appropriate error message is displayed. User remains on login page.	High
TC-AUTH-003	FR-AUTH-001	Login with empty email	Email = blank	1. Leave email blank. 2. Enter password. 3. Click Login.	Email validation message is displayed. Authentication request is not submitted.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-AUTH-004	FR-AUTH-001	Login with empty password	Password = blank	1. Enter valid email. 2. Leave password blank. 3. Click Login.	Password validation message is displayed. Authentication request is not submitted.	High
TC-AUTH-005	FR-AUTH-001	Login with invalid email format	student@	1. Enter invalid email format. 2. Enter password. 3. Submit.	System displays email-format validation and does not submit invalid credentials.	Medium
TC-AUTH-006	FR-AUTH-002	Logout	Active session	1. Open portal. 2. Click Logout.	Firebase session is terminated and user is redirected to login page.	Critical
TC-AUTH-007	FR-AUTH-003	Direct access to protected student page without login	/student/dashboard.html	1. Open protected URL directly.	User is redirected to login/authentication page. Protected data is not displayed.	Critical
TC-AUTH-008	BR-12	Login using deactivated account	Deactivated user credentials	1. Enter credentials. 2. Click Login.	Authentication/access is rejected and user cannot enter the portal.	Critical
TC-AUTH-009	FR-AUTH-004	Password visibility toggle	Valid password	1. Enter password. 2. Click visibility icon.	Password visibility changes without altering the entered password.	Low
TC-AUTH-010	FR-AUTH-005	Session persistence after page refresh	Active session	1. Login. 2. Refresh page.	Authorized session remains valid according to configured authentication/session behavior.	High

2. Role-Based Access Control Test Cases

The platform has four defined roles: **Student, Faculty, HOD/Domain Admin, and Super Admin.**

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-RBAC-001	BR-03	Student accesses Student Portal	Student account	Navigate to student dashboard	Student dashboard loads successfully.	Critical
TC-RBAC-002	BR-03	Student attempts Faculty Portal access	Faculty URL	Enter Faculty Portal URL directly	Access is denied or user is redirected to authorized portal.	Critical
TC-RBAC-003	BR-03	Student attempts HOD Portal access	HOD URL	Enter HOD URL directly	Access is denied.	Critical
TC-RBAC-004	BR-03	Student attempts Super Admin access	Admin URL	Enter Admin URL	Access is denied.	Critical
TC-RBAC-005	BR-05	Faculty accesses assigned academic data	Assigned subject/section	Open assigned subject	Faculty can view/manage authorized academic data.	Critical
TC-RBAC-006	BR-05	Faculty accesses another unauthorized subject	Unassigned subject	Attempt to open/manage unassigned subject	Access is denied and unauthorized data is not exposed.	Critical
TC-RBAC-007	BR-06	HOD accesses own department	Authorized department	Open department data	HOD can access authorized departmental information.	High
TC-RBAC-008	BR-06	HOD accesses another department	Unauthorized department	Attempt to access another department	Access is denied.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-RBAC-009	BR-03	Super Admin accesses administrative functions	Admin account	Open admin portal	Authorized administrative functions are available.	Critical
TC-RBAC-010	BR-17	User attempts unauthorized modification	Record outside authorization scope	Modify record through UI/API	Modification is rejected and unauthorized data remains unchanged.	Critical

3. Student Portal Test Cases

The BRD defines student capabilities including dashboard, academic information, courses, resources, assignments, examinations, attendance, results, productivity tools, calendar, communication, online classes, notifications, AI support, study groups, exam-focused study, and profile/security.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STU-001	F01	Load student dashboard	Student account with academic data	Open Dashboard	Dashboard loads personalized academic information for the student.	High
TC-STU-002	F01	Verify dashboard student identity	Student profile	Open dashboard	Correct student name/profile information is displayed.	High
TC-STU-003	F02	View academic section	Program/semester/section	Open Academic Section	Student's authorized academic information is displayed.	High
TC-STU-004	F03	View courses/subjects	Subject list	Open Courses	Only subjects associated with student's academic context are displayed.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STU-005	F04	View study resources	PDF/document	Open Study Resources	Available authorized learning materials are displayed.	High
TC-STU-006	F04	Download learning resource	Valid resource	Select Download	Authorized file downloads successfully.	Medium
TC-STU-007	F05	View assignment list	Assignment records	Open Assignments	Relevant assignments and their status/deadlines are displayed.	High
TC-STU-008	F08	View examination schedule	Exam data	Open Exams	Correct authorized examination schedule is displayed.	High
TC-STU-009	F09	View attendance	Attendance data	Open Attendance	Student can view attendance applicable to their academic records.	High
TC-STU-010	F10	View published result	Published result	Open Results	Published result is displayed correctly.	Critical
TC-STU-011	F10 / BR-09	View unpublished result	Unpublished result	Open Results	Result is not displayed to the student.	Critical
TC-STU-012	F12	Create personal task	Task title/date	Create task	Task is created and appears in student's task list.	Medium
TC-STU-013	F13	View academic calendar	Event/class/exam	Open Calendar	Relevant academic events are displayed according to student's scope.	Medium

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STU-014	F16	View notifications	Notification record	Open Notifications	Student can view relevant notifications.	High
TC-STU-015	F20	View profile	Student profile	Open Profile	Authorized profile information is displayed.	Medium

4. Faculty Portal Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-FAC-001	F21	Faculty dashboard loads	Faculty account	Open Dashboard	Faculty dashboard loads with authorized academic information.	High
TC-FAC-002	F22	View academic mapping	Subject/section mapping	Open Academic Mapping	Assigned subjects, sections and relevant academic mappings are displayed.	High
TC-FAC-003	F23	Upload learning material	Valid PDF/document	1. Open Materials. 2. Select subject. 3. Upload file.	File is uploaded and associated with the correct academic context.	High
TC-FAC-004	F23	Upload unsupported material	Unsupported file	Upload file	Upload is rejected with validation message.	High
TC-FAC-005	F24	Create assignment	Title, description, deadline	Create assignment	Assignment is created with correct subject and deadline.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-FAC-006	F24	Publish assignment	Draft assignment	Select Publish	Assignment status changes to published and becomes available to authorized students.	Critical
TC-FAC-007	F25	View student submission	Student submission	Open submissions	Faculty can view submissions belonging to authorized students.	High
TC-FAC-008	F25	Evaluate assignment	Marks + feedback	Enter marks and feedback	Evaluation is saved against the correct submission.	Critical
TC-FAC-009	F26	Record attendance	Student attendance	Mark students present/absent	Attendance is saved against correct class/date/student records.	Critical
TC-FAC-010	F27	Enter examination marks	Valid marks	Enter marks and save	Marks are stored against correct student and assessment.	Critical
TC-FAC-011	F27	Enter invalid marks	Negative/over-limit marks	Enter invalid marks	System rejects invalid marks and displays validation message.	High
TC-FAC-012	F28	Schedule online meeting	Meeting details	Create online session	Meeting is created and associated with correct academic context.	High
TC-FAC-013	F29	Send faculty communication	Message	Compose and send message	Message is delivered to authorized recipients.	Medium

5. HOD / Domain Admin Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-HOD-001	F30	HOD dashboard loads	HOD account	Open dashboard	Dashboard displays authorized departmental information.	High
TC-HOD-002	F31	Allocate faculty to subject	Faculty + subject	Select faculty and assign subject	Faculty allocation is saved successfully.	Critical
TC-HOD-003	F31	Allocate unauthorized faculty	Faculty outside scope	Attempt assignment	System prevents unauthorized allocation.	Critical
TC-HOD-004	F32	Monitor students	Department students	Open student monitoring	HOD can view students belonging to authorized department.	High
TC-HOD-005	F33	Monitor faculty	Department faculty	Open faculty monitoring	HOD can view authorized faculty information.	High
TC-HOD-006	F34	Create/update timetable	Class schedule	Add/update timetable entry	Timetable is saved and reflected for authorized users.	High
TC-HOD-007	F35	Approve academic request	Request record	Review and approve	Request status changes to approved and appropriate users are notified if configured.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-HOD-008	F36	Generate department report	Report parameters	Select report criteria	System generates report containing authorized departmental data.	Medium

6. Super Admin Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ADM-001	AD-01	Create student account	Student basic information	Open User Management → Create Student → enter details → Save	Student account is created with correct academic mapping.	Critical
TC-ADM-002	AD-01	Create faculty account	Faculty details	Create Faculty → enter details → Save	Faculty account is created successfully.	Critical
TC-ADM-003	AD-01	Deactivate user	Existing user	Open user → Deactivate	User is marked inactive and cannot authenticate afterward.	Critical
TC-ADM-004	AD-02	Create department	Department name	Add Department → Save	Department is created successfully.	High
TC-ADM-005	AD-02	Create program	B.Tech/BBA/MBA	Add Program → select department → Save	Program is created under correct department.	High
TC-ADM-006	AD-02	Create semester	Semester data	Add Semester → Save	Semester is created under correct program.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ADM-007	AD-02	Create section	Section data	Add Section → Save	Section is created under correct academic hierarchy.	High
TC-ADM-008	AD-02	Create subject	Subject details	Add Subject → Save	Subject is created and associated with correct academic structure.	High
TC-ADM-009	AD-03	Assign user role	Student/Faculty/HOD/Adm in role	Select user → Assign role	Correct role is assigned and corresponding permissions become effective.	Critical
TC-ADM-010	AD-06	View audit logs	Critical action	Open Audit Logs	Authorized administrator can view relevant audit entries.	High

7. Student Account & Credential Test Cases

The platform requirement is that users are provisioned by authorized administrators rather than through unrestricted public registration.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-USER-001	BR-02	Create student using valid information	Student name, college email, program, semester, section	Fill student form and submit	Student account is created successfully.	Critical
TC-USER-002	BR-02	Create student using duplicate email	Existing email	Enter duplicate email and submit	System rejects duplicate email and does not create duplicate account.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-USER-003	BR-02	Create student with missing required information	Missing required field	Submit form	Required-field validation is displayed and account is not created.	High
TC-USER-004	BR-02	Temporary password generation	Valid student data	Create account	System generates/provisions a temporary credential according to configured process.	Critical
TC-USER-005	BR-02	Credential email delivery	Student email	Create student	Student receives required account information through configured email mechanism.	High
TC-USER-006	BR-02	Credential SMS delivery	Student mobile number	Create student	Student receives required account information through configured SMS mechanism.	High
TC-USER-007	BR-02	First login with temporary password	Temporary credential	Login using temporary credential	User can authenticate according to configured first-login flow.	Critical
TC-USER-008	BR-02	Change temporary password	New password	Enter new password and save	Temporary credential is replaced according to password policy.	Critical
TC-USER-009	BR-01	Student attempts public registration	Registration URL	Attempt to create account independently	Unrestricted public account creation is not available.	Critical

Important: The exact password policy, SMS provider, email provider, and retry behavior should be recorded as open requirements if they are not defined in the BRD/SRS rather than invented.

8. Assignment Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ASG-001	F24	Faculty creates assignment	Assignment title, description, deadline	Enter assignment details → Save	Assignment is created successfully.	Critical
TC-ASG-002	F24	Create assignment without title	Blank title	Submit assignment	System displays required-field validation.	High
TC-ASG-003	F24	Publish assignment	Valid assignment	Click Publish	Assignment becomes visible to authorized students.	Critical
TC-ASG-004	F05	Student views published assignment	Assignment	Open Assignments	Assignment details and deadline are displayed.	High
TC-ASG-005	F05	Student submits assignment	Valid file/submission	Open assignment → Upload → Submit	Submission is saved successfully and status changes appropriately.	Critical
TC-ASG-006	F05	Student submits duplicate submission	Second submission	Submit again	System follows configured resubmission rule and prevents unintended duplicate records.	High
TC-ASG-007	BR-16	Student submits after deadline	Late submission	Attempt submission	System applies the configured late-submission rule and displays appropriate status/message.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ASG-008	F25	Faculty evaluates submission	Marks + feedback	Open submission → Enter evaluation → Save	Marks and feedback are stored correctly.	Critical

TC-ASG-009	F25	Student views faculty feedback	Feedback	Open evaluated assignment	Student can view authorized feedback and marks.	High
------------	-----	--------------------------------	----------	---------------------------	---	------

9. Attendance Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-ATT-001	F26	Faculty records attendance	Student attendance list	Open class → Mark attendance → Save	Attendance is stored against correct students/date/class.	Critical
TC-ATT-002	F09	Student views attendance	Student record	Open Attendance	Correct attendance information is displayed.	High
TC-ATT-003	BR-17	Student attempts to modify attendance	Attendance record	Attempt edit	Modification is rejected.	Critical
TC-ATT-004	BR-05	Faculty attempts unauthorized attendance modification	Unauthorized class	Attempt update	System denies modification.	Critical
TC-ATT-005	F26	Invalid attendance data	Invalid/missing attendance value	Submit invalid record	System validates and rejects invalid attendance data.	High

10. Examination & Results Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-EXAM-001	F08	Create examination	Exam details	Enter exam details → Save	Examination record is created.	High
TC-EXAM-002	F08	View examination schedule	Exam schedule	Student opens Exams	Correct examination schedule is displayed.	High
TC-EXAM-003	F27	Enter valid marks	Valid marks	Enter marks → Save	Marks are saved successfully.	Critical
TC-EXAM-004	F27	Enter invalid marks	Marks outside allowed range	Save marks	Invalid marks are rejected.	Critical
TC-RESULT-001	F10	Publish result	Result	Authorized user selects Publish	Result becomes available according to publication rules.	Critical
TC-RESULT-002	F10	Student views published result	Student account	Open Results	Correct result is displayed.	Critical
TC-RESULT-003	BR-09	Student views unpublished result	Unpublished result	Open Results	Result remains hidden.	Critical
TC-RESULT-004	BR-17	Student attempts to modify result	Result record	Attempt modification	Modification is denied.	Critical

11. AI Doubt Solver Test Cases

The platform includes AI-assisted academic support, and your current backend also has a server-side AI endpoint for academic doubt resolution. The backend source shows `POST /api/ai/solve` and keeps the Gemini API key server-side.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-AI-001	F17	Ask valid academic question	"Explain binary search."	Enter question → Submit	AI response is returned and displayed to student.	High
TC-AI-002	F17	Submit empty question	Blank question	Click Submit	Validation message appears and AI request is not sent.	Medium
TC-AI-003	F17	AI service unavailable	Valid question	Submit question	User receives controlled error message instead of application crash.	High
TC-AI-004	F17	Backend AI endpoint unavailable	Valid question	Submit question	Frontend handles failure gracefully and displays appropriate error.	Critical
TC-AI-005	BR-11	Unauthorized AI request	API request	Send request without valid authorization	Request is rejected.	Critical
TC-AI-006	F17	Ask follow-up question	Follow-up question	Ask first question → ask follow-up	System maintains the supported conversation context and returns response.	Medium
TC-AI-007	F17	Escalate unresolved doubt	Unresolved doubt	Select Escalate to Faculty	Doubt is submitted to appropriate faculty workflow.	High

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-AI-008	F14/F17	Faculty responds to escalated doubt	Faculty response	Faculty opens doubt → Respond	Response is saved and made available to authorized student.	High

TC-AI-009	BR-11	Verify AI API key is not exposed	Browser DevTools/network rk	Inspect frontend source and network requests	Secret API key is not exposed to client-side code.	Critical
-----------	-------	----------------------------------	-----------------------------	--	--	----------

12. Study Group Test Cases

The BRD requirement is a maximum of **5 students per study group**.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STUDY-001	F18	Create study group	Group name	Create group	Study group is created successfully.	High
TC-STUDY-002	F18	Join study group	Student account	Open group → Join	Student becomes a group member.	High
TC-STUDY-003	BR-07	Add fifth member	Fifth student	Fifth student joins	Fifth student is successfully added. Total = 5.	Critical
TC-STUDY-004	BR-08	Add sixth member	Sixth student	Sixth student attempts to join	Join request is rejected. Total remains 5.	Critical
TC-STUDY-005	BR-07	Duplicate group membership	Same student	Attempt Join again	Duplicate membership is prevented.	High
TC-STUDY-006	F18	Leave study group	Existing membership	Select Leave	Student is removed from group.	Medium

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-STUDY-007	BR-08	Add member after vacancy	New student	New student joins	New student can join and total does not exceed 5.	High
TC-STUDY-008	BR-03	Unauthorized group access	Private group	Attempt to access restricted group	Unauthorized access is denied.	High

Important boundary case: test concurrent sixth-member attempts as well. The database/security logic must prevent the final membership count from becoming 6 even if two users attempt to join simultaneously.

13. File Upload Test Cases

Your architecture uses Cloudinary for file storage, while application metadata is associated with the platform's academic records.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-FILE-001	BR-16	Upload valid file	Valid PDF	Select file → Upload	File is validated and uploaded successfully.	High
TC-FILE-002	BR-16	Upload invalid file type	Unsupported file	Select unsupported file → Upload	File is rejected with validation message.	High
TC-FILE-003	BR-16	File upload failure	Valid file	Attempt upload	Failure is handled gracefully; incomplete record is not presented as successfully uploaded.	High
TC-FILE-004	BR-03	Unauthorized file access	Restricted file	Attempt direct access	Unauthorized user cannot access file.	Critical

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-FILE-005	BR-16	Verify file metadata	File name/type/academic mapping	Inspect stored metadata	Metadata corresponds to correct file and academic context.	High
TC-FILE-006	BR-16	Upload oversized file	File above configured limit	Upload	Upload is rejected.	High

For **TC-FILE-006**, the exact maximum file size should be taken from the SRS/technical specification. If it has not been defined, record it as an open issue rather than inventing a value.

14. Notification Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-NOTIF-001	F16	Assignment notification	Published assignment	Publish assignment	Relevant student receives notification according to configured notification mechanism.	High
TC-NOTIF-002	F16	Examination notification	Exam event	Create/publish exam	Relevant users receive notification.	High
TC-NOTIF-003	F16	Result notification	Result event	Publish result	Student receives configured notification.	High
TC-NOTIF-004	F17	Faculty response notification	Faculty response	Faculty responds	Student receives response notification if configured.	Medium

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
--------------	----------------	---------------	-----------	------------	-----------------	----------

TC-NOTIF-005	F16	Mark notification as read	Notification	Open notification	Notification status changes to read.	Low
--------------	-----	---------------------------	--------------	-------------------	--------------------------------------	-----

15. Google Meet / Online Class Test Cases

The BRD includes online classes and faculty meeting scheduling.

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
--------------	----------------	---------------	-----------	------------	-----------------	----------

TC-MEET-001	F28	Faculty schedules online class	Subject, date, time, meeting details	Create online session	Online class information is saved successfully.	High
-------------	-----	--------------------------------	--------------------------------------	-----------------------	---	------

TC-MEET-002	F15	Student views online class	Student account	Open class schedule	Student can view authorized online class details.	High
-------------	-----	----------------------------	-----------------	---------------------	---	------

TC-MEET-003	F15	Student opens meeting	Meeting link	Click Join	Student is directed to the configured online meeting.	High
-------------	-----	-----------------------	--------------	------------	---	------

TC-MEET-004	BR-03	Unauthorized user accesses meeting	Meeting URL	Attempt access	Application does not expose meeting information to unauthorized users.	Critical
-------------	-------	------------------------------------	-------------	----------------	--	----------

16. Security Test Cases

Test Case ID	Requirement ID	Test Scenario	Test Data	Test Steps	Expected Result	Priority
TC-SEC-001	BR-14	Unauthorized Firestore read	Restricted document	Attempt read	Firestore Security Rules reject unauthorized read.	Critical
TC-SEC-002	BR-14	Unauthorized Firestore write	Restricted document	Attempt write	Write operation is rejected.	Critical
TC-SEC-003	BR-17	Student modifies marks	Result record	Attempt write/update	Request is rejected.	Critical
TC-SEC-004	BR-17	Student modifies attendance	Attendance record	Attempt write/update	Request is rejected.	Critical
TC-SEC-005	BR-05	Faculty accesses another department	Other department data	Attempt read	Access is denied.	Critical
TC-SEC-006	BR-06	HOD accesses another department	Other department data	Attempt read	Access is denied.	Critical
TC-SEC-007	BR-12	Deactivated user accesses portal	Existing credentials/session	Attempt login/access	Access is blocked.	Critical
TC-SEC-008	BR-11	AI API secret exposure	Gemini/API configuration	Inspect client source/network	Server-side secret is not exposed.	Critical
TC-SEC-009	BR-03	Direct URL manipulation	Unauthorized route	Change URL manually	Unauthorized route is blocked.	Critical
TC-SEC-010	BR-15	Critical action audit	User creation/role change	Perform critical action → inspect logs	Action generates an appropriate audit record.	High

17. Business Rule Test Cases

The BRD explicitly defines business rules BR-01 through BR-18, so these should have a dedicated test suite rather than being tested only indirectly.

Test Case ID	Business Rule	Test Scenario	Test Steps	Expected Result	Priority
TC-BR-001	BR-01	Public self-registration	Open registration route as unauthenticated user	Unrestricted public registration is not provided.	Critical
TC-BR-002	BR-02	Unauthorized user creates account	Attempt account creation without admin authorization	Account creation is rejected.	Critical
TC-BR-003	BR-03	User accesses unauthorized data	Attempt access outside role/scope	Access is denied.	Critical
TC-BR-004	BR-04	Student accesses another academic context	Change student academic identifier/context	Unauthorized academic information remains inaccessible.	Critical
TC-BR-005	BR-05	Faculty accesses unassigned responsibility	Attempt access to unassigned class/subject	Access is denied.	Critical
TC-BR-006	BR-06	HOD accesses another department	Attempt access to another department	Access is denied.	Critical
TC-BR-007	BR-07	Study group reaches five members	Add members sequentially	Fifth member is allowed; group contains maximum 5.	Critical
TC-BR-008	BR-08	Sixth member joins full group	Attempt sixth membership	Sixth member is rejected.	Critical
TC-BR-009	BR-09	Unpublished result visibility	Open unpublished result as student	Result remains hidden.	Critical
TC-BR-010	BR-10	Unauthorized academic structure modification	Unauthorized role modifies program/semester/section	Modification is rejected.	Critical

Test Case ID	Business Rule	Test Scenario	Test Steps	Expected Result	Priority
TC-BR-011	BR-11	AI request without controlled backend authorization	Attempt direct unauthorized AI request	Request is rejected and secret credentials remain protected.	Critical
TC-BR-012	BR-12	Deactivated account login	Attempt login after deactivation	Authentication/access is blocked.	Critical
TC-BR-013	BR-13	Verify deactivated records	Deactivate user → inspect authorized records	Required records are retained according to audit/institutional retention requirements.	High
TC-BR-014	BR-14	Sensitive academic data exposure	Attempt unauthorized API/database access	Sensitive data is not exposed.	Critical
TC-BR-015	BR-15	Audit critical action	Perform critical administrative action	Appropriate audit entry is generated.	High
TC-BR-016	BR-16	File validation	Upload valid/invalid files	Files are validated before availability.	High
TC-BR-017	BR-17	Unauthorized modification	Modify record outside user's scope	Modification is rejected.	Critical
TC-BR-018	BR-18	Academic relationship integrity	Create Institution → Department → Program → Semester → Section → Subject mapping	Relationships are maintained correctly and invalid mappings are rejected.	Critical

18. Negative Test Cases

Test Case ID	Module	Negative Scenario	Test Steps	Expected Result	Priority
TC-NEG-001	Login	Incorrect password	Enter incorrect password	Authentication fails without exposing sensitive information.	High

Test Case ID	Module	Negative Scenario	Test Steps	Expected Result	Priority
TC-NEG-002	Login	Empty required fields	Submit empty form	Validation messages appear.	Medium
TC-NEG-003	User Management	Duplicate email	Create user with existing email	Duplicate account is rejected.	High
TC-NEG-004	Assignment	Missing assignment title	Submit assignment without title	Validation prevents submission.	High
TC-NEG-005	Assignment	Invalid deadline	Enter invalid deadline	System rejects invalid date/time according to configured rule.	High
TC-NEG-006	File Upload	Unsupported file	Upload unsupported format	Upload is rejected.	High
TC-NEG-007	Attendance	Invalid attendance value	Submit invalid attendance	Record is rejected.	High
TC-NEG-008	Examination	Invalid marks	Enter invalid marks	Marks are rejected.	Critical
TC-NEG-009	Study Group	Sixth member	Join full group	Membership rejected.	Critical
TC-NEG-010	AI	AI backend unavailable	Submit question during outage	User receives controlled error.	High
TC-NEG-011	Authorization	Unauthorized URL	Enter restricted URL	Access denied/redirected.	Critical
TC-NEG-012	Database	Unauthorized write	Attempt direct write	Security rules reject operation.	Critical

19. Boundary Test Cases

Test Case ID	Requirement	Boundary	Test Data	Expected Result	Priority
TC-BND-001	BR-07	Study group below maximum	4 members	Fifth member can join.	High
TC-BND-002	BR-07	Study group maximum	5 members	Group remains valid at 5.	Critical
TC-BND-003	BR-08	Above maximum	6th member	Sixth member cannot join.	Critical
TC-BND-004	Assignment	Before deadline	Submission shortly before deadline	Submission follows normal submission rule.	High
TC-BND-005	Assignment	At deadline	Submission at configured deadline	Behavior follows defined deadline rule.	High
TC-BND-006	Assignment	After deadline	Submission after deadline	System applies configured late-submission behavior.	High
TC-BND-007	File Upload	At configured file-size limit	File exactly at limit	Upload behavior follows configured limit.	Medium
TC-BND-008	File Upload	Above configured limit	File larger than limit	Upload is rejected.	High
TC-BND-009	Required Fields	Minimum valid input	Minimum permitted values	Record is accepted if all requirements are satisfied.	Medium
TC-BND-010	Required Fields	Missing required value	Blank required field	Validation prevents submission.	High

20. Integration Test Cases

Test Case ID	Integration	Test Scenario	Test Steps	Expected Result	Priority
TC-INT-001	Firebase Authentication	Login integration	Login with valid account	Firebase authenticates user and application establishes authorized session.	Critical
TC-INT-002	Firestore	User profile retrieval	Login → Dashboard	Correct Firestore user data is retrieved.	Critical
TC-INT-003	Firestore	Assignment persistence	Create assignment	Assignment is persisted and retrievable.	Critical
TC-INT-004	Cloud Functions	AI integration	Submit AI question	Request reaches controlled backend and response is returned.	Critical
TC-INT-005	Cloudinary	File integration	Upload academic file	File is uploaded and metadata is associated correctly.	High
TC-INT-006	FCM	Notification integration	Trigger notification event	Notification is delivered according to configured behavior.	High
TC-INT-007	Google Meet	Meeting integration	Faculty creates online session	Meeting information is stored and available to authorized students.	High

21. End-to-End Test Cases

Test Case ID	E2E Scenario	Test Steps	Expected Result	Priority
TC-E2E-001	Student onboarding	Create student → provision credentials → student logs in → open dashboard	Student account is created, credentials delivered through configured process, and student reaches authorized dashboard.	Critical
TC-E2E-002	Assignment lifecycle	Faculty creates assignment → publishes → student views → submits → faculty evaluates → student views feedback	Complete assignment workflow works correctly.	Critical
TC-E2E-003	AI doubt lifecycle	Student asks AI → AI responds → unresolved doubt escalated → faculty responds	Complete doubt-resolution workflow works correctly.	High
TC-E2E-004	Study group lifecycle	Create group → students join → fifth joins → sixth attempts → member leaves → new member joins	Capacity rule remains enforced throughout lifecycle.	Critical
TC-E2E-005	Academic structure lifecycle	Create department → program → semester → section → subject → faculty/student mapping	Academic hierarchy is created and relationships remain consistent.	Critical

22. Non-Functional Test Cases

Performance

Test Case ID	Requirement	Test Scenario	Test Steps	Expected Result	Priority
TC-PERF-001	NFR-PERF	Login response	Perform valid login under normal test environment	Response meets the performance target defined in SRS.	High
TC-PERF-002	NFR-PERF	Dashboard loading	Open student dashboard	Dashboard loads within the defined target.	High
TC-PERF-003	NFR-PERF	Assignment retrieval	Open assignment list with representative data	Assignment list loads within defined target.	High
TC-PERF-004	NFR-PERF	Concurrent access	Simulate expected concurrent users	System remains functional within defined performance limits.	High

The actual numeric performance target should come from the SRS. It should not be invented in the test document.

Responsive Design

Test Case ID	Test Scenario	Test Steps	Expected Result	Priority
TC-RESP-001	Desktop student portal	Open student dashboard on desktop browser	Layout displays correctly without clipping or overlap.	Medium
TC-RESP-002	Tablet student portal	Open portal on tablet viewport	Navigation and content remain usable.	Medium
TC-RESP-003	Mobile student portal	Open portal on mobile viewport	Responsive layout is usable and content is accessible.	High
TC-RESP-004	Faculty portal responsive behavior	Open faculty pages at different viewport sizes	Faculty pages remain usable without layout-breaking issues.	Medium

Accessibility

Test Case ID	Test Scenario	Test Steps	Expected Result	Priority
TC-ACC-001	Keyboard navigation	Navigate login using keyboard only	All important controls can be reached and operated.	Medium
TC-ACC-002	Form labels	Inspect login/user forms	Inputs have meaningful labels or accessible names.	Medium
TC-ACC-003	Error messages	Submit invalid form	Validation errors are understandable and associated with relevant fields.	Medium
TC-ACC-004	Focus visibility	Navigate controls using keyboard	Focus state remains visible.	Low

23. Test Execution Summary

Since these are **test cases being prepared before execution**, use:

Metric	Initial Value
Total Test Cases	To be calculated from final approved suite
Executed	0
Passed	0
Failed	0
Blocked	0
Not Executed	Total Test Cases
Pass Percentage	N/A
Fail Percentage	N/A

The source test-document requirements explicitly say not to fabricate execution results.

24. Test Case Status Values

Use:

Status	Meaning
Not Executed	Test case has been documented but not run
Passed	Actual result matches expected result
Failed	Actual result differs from expected result
Blocked	Test cannot be executed because of a dependency/blocker
Not Applicable	Requirement/test is formally determined not applicable

25. Priority Definitions

Priority	Definition
Critical	Failure prevents core system usage or creates serious security/data-integrity risk
High	Major feature or important workflow is unavailable or incorrect
Medium	Feature works incorrectly but a workaround exists
Low	Minor issue with limited functional impact

These definitions are also specified in your test-document source.

Sign-Off

Role	Name	Status	Date	Signature
Tech Lead	Riddhi Zunjarrao			
Solution Architect	Ragini Singh			
Project Manager / BA	Sakshi Shingole			
Scrum Master	Sweta Kadam			